R4Tun Combined Grouped Review (Deduplicated)

Source basis:

- `r4tun_final_manuscript_sentence_review.md`
- `r4tun_review_v4_pre_submission_review.md`

Scope: deduplicated, grouped, non-redundant issue list that preserves all unique points raised across both reviews.

1) Numerical And Consistency Issues (Highest Priority)

1. Abstract OA clause is grammatically incomplete

Original

> ... increased mean mIoU by at least 110.7% and 26.8% Overall Accuracy (OA).

Issue

Reads as if OA itself is 26.8% rather than OA improvement.

Suggested revision

> ... increased mean mIoU by 110.7%–128.0% and overall accuracy (OA) by 26.8%–34.1% relative to the static SAM4Tun baseline.

2. Relative-improvement ranges conflict across sections

Issue

Conclusions reports values recomputed from rounded numbers, while Highlights/Abstract/Results use unrounded values.

Conflicts to align

- Overall: `113.3%–126.7%` vs `110.7%–128.0%`
- Regular: `+72.4%–86.2%` vs `+70.0%–83.8%`
- Complex: `+350.0%–375.0%` vs `+323.1%–361.7%`

Suggested revision

Use one canonical set everywhere (preferably Section 4.1 values from unrounded means).

3. Four-stage description conflicts with five API calls

Original

- Four stage agents are described in methodology.
- Appendix reports `LLM API calls per tunnel: 5` and totals based on 5.

Issue

Either an unmentioned stage exists (e.g., detecting), or API-call counts are incorrect.

Suggested revision

- If four adapted stages: `4 calls per tunnel`, `120 per condition per LLM`.
- If five adapted stages: explicitly add the fifth stage in architecture/method sections and figures.

4. Reference diameter definitions are inconsistent (5.5 / 5.60 / 5.32)

Issue

Nominal and estimated diameters are mixed without explicit definition, which also causes `+34%` vs `+39%` confusion.

Suggested revision

Define once:

- Nominal diameter = design value (e.g., 5.60, 7.50)
- Estimated diameter = measured from point-cloud processing (e.g., 5.32, 7.41)

5. Parameter-count drift ("approximately 80" vs "81")

Issue

TeX and PDF differ.

Suggested revision

Use one exact count and cross-check against appendix parameter tables.

6. "Default API configuration" conflicts with `Temperature = 0`

Issue

Temperature zero is an explicit override.

Suggested revision

> Other than setting temperature to 0, all other API settings used vendor defaults.

7. Citation [50] does not support all listed models

Issue

[50] is GPT-4o system card, while table lists Opus-4.6, GPT-5.4, Gemini-3-Flash.

Suggested revision

Remove [50] from that sentence, or add correct per-vendor references.

8. Terminology and formatting drift

Issue

Inconsistent use of:

- `Non-LLM` vs `non-LLM`
- `tunnel family` vs `tunnel category`
- `regular-category` vs `regular category`

- `95 %` vs `95%`

Suggested revision

Standardize globally (recommended: `non-LLM`, `tunnel category`, noun form without hyphen, compact `%`).

2) Overstatements To Soften

9. "strong adaptability" and similar broad claims

Original

> Existing tunnel-segmentation approaches do not jointly provide strong adaptability...

Issue

Too categorical/subjective.

Suggested revision

> ...have not yet demonstrated, in a single workflow, both robust adaptation to changed conditions and an auditable adaptation process.

10. "self-verifying intermediate steps"

Issue

Overstates reliability and conflicts with later caveat that constraints are not guaranteed.

Suggested revision

Use "prompted consistency checks" and acknowledge probabilistic nature.

11. "objective tunnel conditions" in cross-model analysis

Issue

Cannot isolate "objective condition" causality from shared prompt/knowledge/model priors.

Suggested revision

Phrase as consistency under same inputs/prompt structure.

12. "context design is shown to help"

Issue

Too assertive for evidence level.

Suggested revision

Use "ablation results indicate/suggest".

13. LLM-vs-rule conclusion hedging is inconsistent

Issue

Some sections use cautious wording ("consistent with additional contribution"), others sound definitive.

Suggested revision

Use cautious causal language throughout.

14. Practical-impact sentence sounds promotional

Original

"reallocates expert effort", "one-off authoring step", "affordable APIs".

Issue

Reads as promotional and overgeneralized.

Suggested revision

Use "may shift" language; call API costs "indicative under tested settings".

3) Internal Logic / Reasoning Consistency

15. "strictly bounded" conflicts with "does not guarantee constraints"

Issue

Deterministic wording clashes with non-deterministic implementation.

Suggested revision

Replace "strictly bounded" with "instructed to stay within" and clarify non-guaranteed enforcement.

16. Convergence sentence is internally mixed

Original

Claims rule+LLM "converge" yet "LLMs still score higher" in same long sentence.

Suggested revision

Split into two sentences:

- Both remain low in absolute performance on complex category.
- LLM still has small mean advantage.

4) AI-Like / Vague / Promotional Phrasing

17. Replace vague or hype-like wording

Target phrases:

- "handle varying tunnel conditions"
- "degrades silently"
- "diagnostic feedback"
- "explicit rationale"
- "leverage ... foundation models"
- "actual geometric reality"
- "the agent attempts to..."

Suggested style

Use concrete, observable language tied to measured effects and pipeline behavior.

18. Circular wording in dataset description

Original

> complex interleaved key-block arrangement

Issue

"complex" for complex category is circular.

Suggested revision

> interleaved key-block arrangement that does not repeat ring-to-ring.

5) Repetition To Reduce

19. Repeated triplet and near-identical claims

Repeated heavily:

- "memory, state, and knowledge"
- "reference configuration"
- "per-tunnel expert re-tuning/retuning"
- duplicated "state provides quantitative evidence..." sentence
- duplicated per-class IoU summary sentence

Suggested revision

Define once, then use shorthand (`m+s+k`) and cross-references.

20. Keep one strong version of each finding

Examples:

- keep per-class summary once in Results, brief reference in Conclusions.
- keep state-mechanism explanation once in Ablation, brief callback in Discussion.

6) Grammar, Typo, And Formatting Fixes

21. Typo and spacing

- `JSON ouput` -> `JSON output`
- `protocol(Algorithm 1)` -> `protocol (Algorithm 1)`
- `The mean increment are narrow` -> `The mean increments are small`

22. Figure 6 caption grammar

Issue

Missing grammatical subject for "constructs" / "feeds".

Suggested revision

Use a clear subject and two-step sentence structure.

23. Appendix heading grammar

`Non-LLM rules-based` -> `Non-LLM rule-based`.

24. Unit and symbols

- `pts/m3` -> `pts/m³` (or journal-preferred equivalent)
- normalize `%` spacing
- verify math rendering in PDF (`□ar{v}`, summation bounds) is typographically correct.

25. Source artefact in TeX

Remove stray `t d` in the sentence beginning "The pipeline contains...".

7) Structure And Organization Fixes

26. Move metrics/sensitivity out of Non-LLM subsection

Issue

`3.5.1 Evaluation metrics` and `3.5.2 Sensitivity analysis` are global methods, not specific to non-LLM baseline.

Suggested revision

Renumber as standalone methodology subsections.

27. Align Highlights and Contributions

Use identical headline claims and numbers in both places after numeric reconciliation.

28. Verify Algorithm 1 float placement/caption

Ensure caption and body render together and are not detached across columns/pages.

8) Consolidated High-Quality Abstract (Optional Replacement)

> Automated inspection of segmental tunnel linings requires segmentation pipelines that adapt to varying point-cloud geometry, yet expert-tuned pipelines often degrade when tunnel conditions depart from the reference setting. This paper presents R4Tun, a large language model (LLM)-driven adaptation framework that extends an expert-designed pipeline (SAM4Tun) with bounded parameter tuning informed by structured context (memory, state, and knowledge; m+s+k). On 30 selected Seg2Tunnel subsets (13 regular, 17 complex) and three LLMs (Opus-4.6, GPT-5.4, Gemini-3-Flash), the m+s+k design increased mean Intersection-over-Union (mIoU) from 0.150 to 0.316–0.342 (+110.7%–128.0%) and overall accuracy (OA) from 0.411 to 0.522–0.552 (+26.8%–34.1%). On regular tunnels mIoU rose from 0.291 to 0.495–0.535 (+70.0%–83.8%); on complex tunnels mIoU rose from 0.042 to 0.178–0.194 (+323.1%–361.7%). Across 270 adapted runs (30 tunnels × 3 LLMs × 3 context settings), the three LLMs followed similar parameter-adjustment trends and consistently selected a shared set of critical parameters. These results suggest that LLM-driven adaptation supplied with structured context can improve transfer of a fixed segmentation pipeline across changed

tunnel conditions and reduce the need for per-tunnel expert retuning, particularly when target tunnels remain close to the reference configuration.

9) One-Pass Final Checklist

1. Unify all numeric ranges in Highlights/Abstract/Results/Conclusions.
2. Resolve 4-vs-5 stage inconsistency and update architecture, methods, tables.
3. Define nominal vs estimated diameter once and reuse consistently.
4. Standardize terms/capitalization/percent style globally.
5. Replace unsupported or mismatched citations.
6. Soften overstatements; keep causal claims cautious.
7. Remove duplicated wording; keep one canonical statement per finding.
8. Apply typo/grammar/unit/math-format fixes.
9. Reorganize subsection numbering for methodological clarity.
10. Verify final PDF rendering (figures, algorithm floats, equations).